

UNIT II

OPTICS

Interference - Air Wedge - Mechelson's interferometer - Wave length Determination - Interference Filter - Antireflection Coatings. Diffraction - Diffraction grating - Dispersive power of grating - Resolving Power of Grating and Prism Polarisation - Huygens Theory of Double Refraction - Quarter and Half Wave Plates - Specific Rotary Power Laurent Half Shade Polarimeter.

2.1 INTRODUCTION

We know the light has dual nature, it can behave both as a particle and as a wave. The wave nature of light is proved by the phenomenon of interference of light and the phenomenon of polarization helps us to establish that the light waves are transverse in nature. In this chapter we are going to discuss about the interferometry and their application in various fields.

2.2 INTERFERENCE

When two light waves from the sources superimposes on each other then the resultant amplitude in the region of superposition is different than the amplitude of the individual waves. This modification of light i.e. change in amplitude is called as **interference**.

When the amplitude of the resultant is equal to the sum of the amplitude of the two waves, then the interference is called **constructive interference** and the field of view is bright. When the amplitude of the resultant is equal to the differences of two amplitudes the interference is known as **destructive interference** and the field of view is dark.

Using the principle of interference one can design the antireflection coating element used in cameras, telescopes etc. The phenomenon of interference is also used to study the formation of colours in soap bubbles, thin film of petrol, diesel etc.,

2.3 CONDITIONS FOR INTERFERENCE :

For Interference to occur between two sources, the following conditions have to be satisfied.

1. There must be two sources which are coherent in nature (i.e.) They must be in same wavelength, frequency and amplitude. The sources should have constant phase difference (or) zero phase difference.
2. The sources should be closer to each other and the distance between the sources and the screen must be larger.
3. For constructive interference (i.e) for bright fringes to appear, the path difference between the two waves which are interfering should be equal to $n\lambda$

$$\text{Path Difference} = n\lambda$$

4. For destructive interference (i.e) to get dark fringes.

$$\text{Path difference} = (2n+1)\frac{\lambda}{2}$$

2.4. FRINGES PRODUCED BY A WEDGE SHAPED THIN FILMS (AIR WEDGE)

Definition:

Two optically plane glass plates are placed one over the other and tied by means of a rubber band at one end. The given material of wire or paper is introduced on the other end, so that an air wedge is formed

Let us consider two plane surfaces OA and OB inclined at an angle θ , enclosing a wedge shaped air film as shown in Fig. 2.1

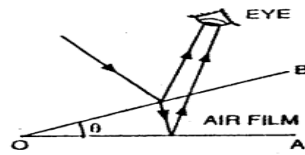


Fig. 2.1

Let μ be the refractive index of the enclosing film in between two plates OA and OB. The thickness of the film increases from O to A. when this thin air film is illuminated by the monochromatic light, interference occurs between the rays reflected from upper and lower surface of the film. The interfering rays do not enter the eye parallel but they appear to diverge from a point near the film.

Let 't' be thickness of the film at a distance x from the edge, We know that the path difference for reflected light $\Delta = 2\mu t \cos r$

For normal incidence

$$r = 0, \cos 0 = 1$$

$$\text{The path difference } \Delta = 2\mu t \dots\dots\dots(1)$$

If ' θ ' is every small then we can write

$$t = x\theta \dots\dots\dots(2)$$

substitute eq.(2) in eq. (1)

$$\text{The path difference } \Delta = 2\mu x \theta \dots\dots\dots (3)$$

The condition for formation of bright fringes due to the reflected light is

$$\Delta = (2n + 1) \frac{\lambda}{2} \dots\dots\dots (4)$$

Substitute eq. (3) in eq. (4)

Here for air $\mu = 1$

$$2x\theta = (2n + 1) \frac{\lambda}{2} \dots\dots\dots (5)$$

similarly, we know the condition for dark fringes due to reflected light is

$$\text{The path difference } \Delta = n\lambda \dots\dots\dots (6)$$

From eq. (3) we can write

$$\text{The path difference } 2\mu x\theta = n\lambda, \text{ if } \mu = 1$$

$$2x\theta = n\lambda \dots\dots\dots (7)$$

2.4.1. FRINGE WIDTH : (β)

It is defined as the distance between any two consecutive bright (or) dark fringes.

2.4.2. EXPRESSION FOR FRINGE WIDTH : (β)

Case (i) : Let us consider the dark fringe from the edge O. Let x_n be the distance of the n^{th} dark fringe of thickness (t_n) and $x_{(n+1)}$ be the distance of the $(n+1)^{\text{th}}$ dark fringe of thickness (t_{n+1}) as shown in the Fig. 2.2

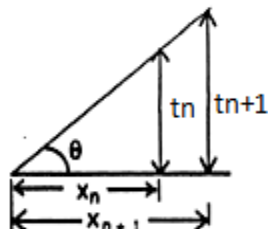


Fig. 2.2

From eq (7) We can write

$$x_n = \frac{n\lambda}{2\theta} \dots\dots\dots (8)$$

$$\text{and } x_{n+1} = \frac{(n+1)\lambda}{2\theta} \dots\dots\dots (9)$$

By definition of fringe width

$$\beta = x_{n+1} - x_n = \frac{(n+1)\lambda}{2\theta} - \frac{n\lambda}{2\theta}$$

$$\text{(or) Fringe width } \beta = \frac{\lambda}{2\theta} \dots\dots\dots (10)$$

Case ii : Similarly if we consider any two consecutive bright fringes, then the fringe width β will be the same From eq. (5) we can write

$$x_n = \frac{\lambda}{4\theta} (2n+1) \text{ and}$$

$$x_{n+1} = \frac{\lambda}{4\theta} (2(n+1)+1) = \frac{\lambda}{4\theta} (2n+3)$$

$$\therefore \beta = x_{n+1} - x_n = \frac{\lambda}{4\theta} (2n+3 - 2n-1)$$

$$\therefore \beta = \frac{\lambda}{2\theta} \dots\dots\dots (11)$$

2.4.3. APPLICATION OF AIR WEDGE

1. To find the thickness of any thin material

Principle

Two plane glass plates are inclined at an angle (θ) by introducing a thin material (e.g. hair), forming a wedge shaped air film. The film is illuminated by sodium light. Interference occurs

between the two rays, one reflected from the front surface and the other by internal reflection at the back surface. Therefore straight line fringes parallel to the edge of the wedge are obtained. Using the theory of air wedge the thickness of the material can be determined.

DESCRIPTION

Two optically plane glass plates are placed one over the other and tied by means of a rubber band at one end. The given material of wire or paper is introduced on the other end, so that an air wedge is formed between the plates as shown in Fig. 2.3.

Experimental set up:

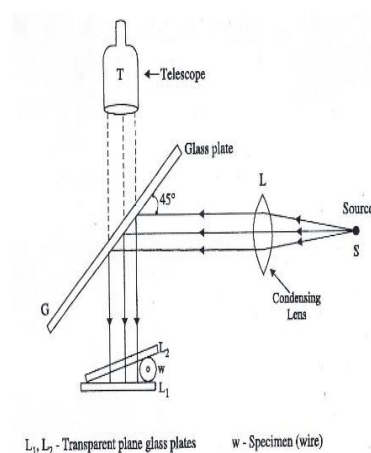


Fig. 2.3

This set up is placed on the horizontal bed plate of the travelling microscope. Light from the source 'S' can be made to fall on the Air Wedge set up with the help of a condensing lens (L) and 45° angled glass plate (G)

WORKING

Light from the sodium vapour lamp (S) is rendered parallel by means of a condensing lens (L). The parallel beam of light is incident on a plane glass plate (G) inclined at an angle of 45° and gets reflected. The reflected light is incident normally on the glass plates in contact.

Interference takes place between the light reflected from the top and bottom surfaces of the glass plates and is viewed through the travelling microscope (M). Since the thickness of the materials, remains constant in the direction parallel to the thin edge of the wedge, a large number of equally spaced dark and bright straight line fringes are formed parallel to the edge of contact as shown in Fig. 2.4.

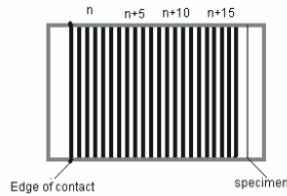


Fig. 2.4

The microscope is adjusted so that the bright (or) dark fringe near the edge of contact is made to coincide with the vertical cross wire and this is taken as the n^{th} fringe. The reading from the horizontal scale of the travelling microscope is noted. The microscope is moved across the fringes using the horizontal transverse screw and the readings are taken when the vertical cross wire coincides with every successive 5 fringes (5, 10, 15, 20.....) The width of every 20 fringes is calculated and the width of one fringe is calculated. The mean of this gives the fringe width (β)

We know from the theory of Air wedge, the fringe width $\beta = \frac{\lambda}{2\theta}$ (1)

If ' l ' is the distance between the edge of contact and the material (say the distance of the last fringe) and ' t ' is the thickness of the material, then

For θ being very small from the Fig. 2.5

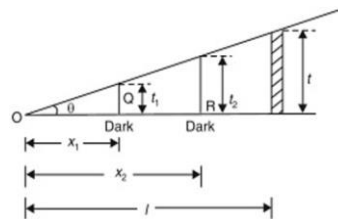


Fig. 2.5

From Fig. 2.5 We can write $\theta = \frac{t}{l}$ (2)

Substituting eq (2) in (1) we get

$$\text{Fringe width } \beta = \frac{\lambda l}{2t}$$

$$\therefore \text{ Thickness of the material } t = \frac{\lambda l}{2\beta}$$

Thus, by finding λ , l and β the thickness of any thin material can be determined using air wedge.

2. Testing of Flatness of a given surface principle

Principle

If two surfaces are perfectly flat then the air film gradually varies in thickness. The fringes will be of equal thickness because each fringe is the locus of the points at which the thickness of the film has constant value. If the fringes are not equal in thickness it means the surfaces are not flat.

Theory :

We know the conditions for the formation of bright fringes due to the reflected light is

$$2\mu t \cos r = (2n+1)\frac{\lambda}{2}$$

For normal incidence $r = 0 \quad \therefore \cos r = 1$

$$\therefore 2\mu t = (2n+1)\frac{\lambda}{2}$$

From the above equation, a fringe will be characterized by a particular value 'n' and if the thickness of the air film or flatness of the surface is constant, we get straight line fringes.

EXPLANATION :

Let us consider two surfaces OA and OB. The surface OB which is to be tested is placed in contact with optically flat glass plate (i.e) surface OA, forming an air wedge as shown in Fig 2.6 (a). The fringes are observed in the field of view. The fringe pattern depends upon the flatness of the surface OA and OB

(i) If the fringes are of equal thickness as shown in Fig. 2.6 (a) then the surface OB is said to be flat.

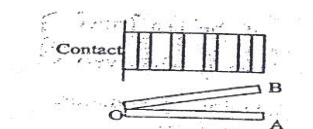


Fig. 2.6 (a)

(ii) If the fringes observed are of unequal thickness it implies that the surface OB is not flat. For this we have two cases.

a) If the fringes are curved towards the contact edge, the surface is concave as shown in Fig. 2.6

(b)

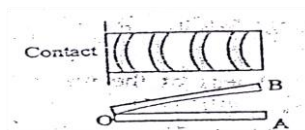


Fig. 2.6 (b)

b) If the fringes are curved away from the contact edge, the surface is convex as shown in Fig. 2.6 (c)

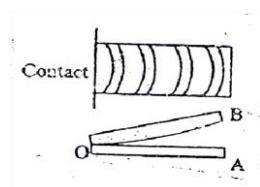


Fig. 2.6 (c)

Thus to make the surface OB to be flat, the surface OB has to be polished and the same process can be repeated until the fringe pattern is obtained as straight lines.

2.5. INTERFEROMETRY

The phenomenon of interference has been used to test the flatness of surface and also to reduce reflecting power of the lens and prisms. Instruments based on the principle of interference of light are known as interferometers. Michelson designed an interferometer to determine the wavelength of the light, resolution of the spectral line and the thickness of the thin transparent materials.

2.6 MICHELSON'S INTERFEROMETER AND TYPES OF FRINGES.

Principle :

The amplitude of light beam from a source is divided into two parts of equal intensities by partial reflection and transmission. These beams are then sent in two directions at right angles and are brought together after they suffer reflection from plane mirrors to produce interference fringes.

Construction:

It consists of two highly polished mirrors M_1 and M_2 which are at right angles to each other. There are two optically flat glass plates G_1 and G_2 of same thickness and made up of same material placed parallel to each other. These plates are inclined at an angle of 45° with the mirror M_1 and M_2 . The plate G_1 is half silvered at the back so that the incident beam divided into two beams viz, reflected beam and transmitted beam of equal intensity. The mirrors M_1 and M_2 are provided with screws on their back, so that they can be adjusted exactly perpendicular to each other.

The two rays which enter the telescope are originally derived from the same single beam; hence they cause the interference fringes in the field to view of the telescope. Hence a path difference can be introduced between the two reflected rays by moving the mirror M_1 . It is clear from the Fig. 2.7 that ray PC passes twice through the glass plate G_1 . i.e. 1st through PQ and 2nd through QP, after reflection from the mirror M_1 , whereas the ray PB does not even passes once through G_1 , even after reflection from the mirror M_2 . Thus in the absence of the glass plate G_2 the path traced by the beam between G_1 M_1 and G_1 M_2 are not equal. To equalize the path difference, a glass plate G_2 of same thickness and materials as that of G_1 is introduced between G_1 and M_2 . So that the ray PB will also pass twice, i.e. 1st through RO in glass plate G_2 and 2nd through OR in Glass plate G_2 after reflection from the mirror M_2 . The mirror M_1 is mounted on a carriage which can be moved forward and backward using the handle 'H'. The distance at which the M_1 is moved can be read with the help of the scale as shown in Fig. 2.7. The interference fringes can be observed in the field of view of the Telescope (T)

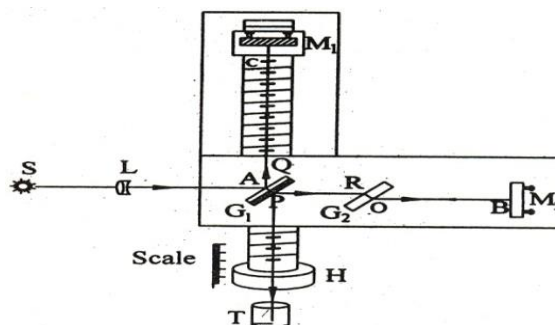


Fig. 2.7

WORKING :

Light from a monochromatic source S is made parallel with the help of collimating lens L. The light beam is allowed to fall on the semi silvered glass plate G_1 . It is partly reflected at the back surface of G_1 and travel towards M_1 i.e. along (AC) and partly transmitted towards M_2 (i.e along AB). These two rays travel along two mutually perpendicular paths and are reflected back by the mirror M_1 and M_2 . These two rays again meet at glass plate G_1 and enter a short focus telescope T. Since the glass plate G_2 is used to compensate the path difference between the two rays, it is called as a **compensation plate**. Thus the path of the two rays viz. PB and PC are made equal.

2.6.1. TYPES OF FRINGES

(i) Circular fringes

When mirror M_2 is exactly perpendicular to mirror M_1 and the virtual mirror M_2' (which is image of M_2) an air film of constant thickness is enclosed between them as shown in Fig. 2.8.

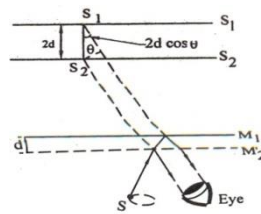


Fig. 2.8.

The air film gives reflected beam to interfere. The path difference depends upon the following factors.

- (i) The separation between M_1 and M_2'
- (ii) The angle suspended on the eye
- (ii) The inclination between M_1 and M_2'

It can be seen from the above Fig.2.8 that the path differences are different for different values of θ . The source S is an extended source and S_1 and S_2 are the virtual source due to M_1 and M_2' . If the distance between M_1 and M_2' is d and the distance between S_1 and S_2 is $2d$, then

The path difference between the two rays = $2d \cos \theta$

The equation for maxima is given by $2d \cos \theta + \frac{\lambda}{2} = n\lambda$, Where $\frac{\lambda}{2}$ is the additional path

difference between the rays

Case (i)

When M_1 and M_2' coincides, the path difference is zero and the field of view is perfectly dark as shown in Fig. 2.9

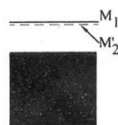


Fig. 2.9

Case (ii)

When M_1 is moved either way parallel to itself, widely spaced circular fringes are produced. The width of the fringes depends upon the path difference between the rays Fig. 2.10.

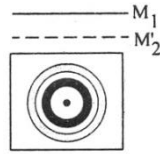


Fig. 2.10

(i) Localised fringes

When the mirror M_2 is not exactly perpendicular to M_1 or if M_1 and the vertical Mirror M_2' are inclined then the air film enclosed between them is wedge shaped. For small path difference, the fringes are nearly straight but for large path difference the fringes are generally curved and are always convex towards the thin edge of the wedge. When the two mirrors M_1 and M_2' intersect in the middle, straight line fringes are observed as shown in Fig. 2.11(a). When the two mirrors are inclined, curved fringes are observed as shown in Fig. 2.11 (b&c)

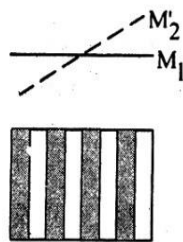


Fig. 2.11 (a)

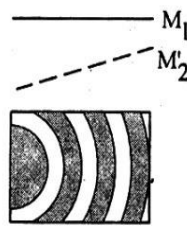


Fig. 2.11 (b)

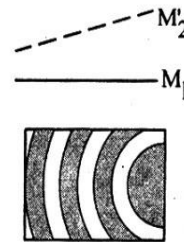


Fig. 2.11 (c)

(ii) Localised white light fringes

With white light, the fringes are observed only when the path difference is very small. The fringe of zero thickness is dark (i.e) central fringe and the other fringes are coloured due to overlapping of various colours. If the film is thick, uniform illumination is observed. White light fringes are useful to find the zero path difference, especially in the standardization of meter.

2.6.2. APPLICATION OF MICHELSON INTERFEROMETER

Michelson interferometer can be used to determine.

- (i) The wavelength of the monochromatic light.
- (ii) The difference between the two neighboring wavelengths or resolution of the spectral lines.
- (iii) The refractive index and thickness of the various thin transparent materials and
- (iv) For the measurement of the standard meter in terms of wave length of light.

2.6.3. DETERMINATION OF WAVE LENGTH OF A MONOCHROMATIC SOURCE OF LIGHT

The mirrors M_1 and M_2 are adjusted to get the circular fringes in the field of view of the telescope. Then the mirror M_1 is moved with the help of handle of the micrometer screw and the number of fringes that cross the field of view is counted. If 'n' is the number of fringes that cross the

field and 'd' is the distance through the mirror is moved, then, from the theory of circular fringes.

$$2d \cos\theta = n\lambda \text{ (Bright)}$$

Since θ is very small and is ≈ 0

$$\therefore 2d = n\lambda$$

$$d = \frac{n\lambda}{2}$$

$$\text{or } \lambda = \frac{2d}{n}$$

If $n = 1$ (i.e) only one fringe crosses the centre of field of view, then $d = \frac{\lambda}{2}$ (i.e) the mirror moves through a distance which is equal to half of the wavelength $\therefore \lambda = 2d$

2.6.4. DETERMINATION OF THICKNESS OF A THIN TRANSPARENT SHEET

Initially, the Michelson interferometer is set for localised fringe of white light. The central fringe in the field of view is made to coincide with the vertical cross wire and the position of the mirror M_1 is noted. Now the thin transparent sheet of known refractive index (μ) is introduced between the glass plate G_2 and the mirror M_2 . Due to the introduction of thin sheet, a path difference is introduced and hence the central fringe is shifted from its original position to some other position.

In this position the path difference will be $2(\mu-1)t$. The position of the mirror M_1 is adjusted until the central fringe again coincides with the cross wire. The distance 'd' moved by the mirror M_1 is noted with the help of the scale.

Therefore, we can write the path difference $2(\mu-1)t = 2d$

$$\text{or } t = \frac{d}{(\mu-1)}$$

Thus, from the above formula the thickness of the thin transparent sheet can be determined.

2.7. ANTIREFLECTION COATING (OR) NON REFLECTING

One of the important applications of the interference phenomenon is the manufacture of the non-reflecting film. We all are familiar about photography. In photography, while we take photographs, the light from the object should fall on the lens system in the camera and should be completely absorbed by those lenses in the camera.

But in practice some lenses are not capable of completely absorbing the light falling over it, so that the resolution of the picture may be very poor. In order to increase the absorbing capacity (or) to minimize the loss of light due to reflection, scattering etc., a coating can be given over the lenses this coating prevents the loss of light and act as an anti-reflecting element and hence known as **anti-reflecting coating**.

The loss in light due to reflection is caused when the light passes through the interface of materials with different refractive indices.

Reflectivity: If μ_g and μ_a are the refractive indices of glass and air respectively, then

$$\text{Reflectivity} = \left[\frac{\mu_g - \mu_a}{\mu_g + \mu_a} \right]^2$$

If $\mu_g = 1.6$ and we know refractive index of air (μ_a) = 1 then

$$\text{Reflectivity} = \left[\frac{1.6-1}{1.6+1} \right]^2 \Rightarrow 0.05325$$

(or) Reflectivity is 0.0532, which implies that 5.32% of the light is reflected from the interface and is lost.

If we reduce the refractive index of the materials then the loss in light due to reflection can be minimized.

2.7.1. SINGLE ANTI - REFLECTION COATINGS

In the case of transparent materials, whenever a beam of light passes from a rarer medium to a denser medium, a certain fraction of the incident light is always reflected by the interface. Due to this, the transmitted light will suffer in intensity and it reduces the contrast of the images. Therefore, a layer of anti - reflection coating has to be given to prevent the loss as follows.

Let us consider a material to be given anti - reflecting coating. Let μ_c be the refractive index of coating material, μ_g be the refractive index of glass and μ_a the refractive index of air as shown in Fig. 2.12

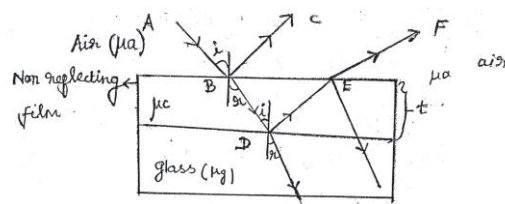


Fig. 2.12

$$\text{Then, reflectivity at air - coating surface} = \left[\frac{\mu_c - \mu_a}{\mu_c + \mu_a} \right]^2 \dots\dots\dots (1)$$

$$\text{Then, reflectivity at air - glass surface} = \left[\frac{\mu_g - \mu_c}{\mu_g + \mu_c} \right]^2 \dots\dots\dots (2)$$

The reflected rays BC and DF should have constant phase difference of π (or) path difference of $\frac{\lambda}{4}$ then only the output beam will appear dark and the amplitude of the rays i.e. reflectivity should be equal.

$$\therefore (1) = (2) \Rightarrow \left[\frac{\mu_c - \mu_a}{\mu_c + \mu_a} \right]^2 = \left[\frac{\mu_g - \mu_c}{\mu_g + \mu_c} \right]^2 \dots\dots\dots (3)$$

Since refractive index of air = $\mu_a = 1$

Solving equation (3) we can write = $\mu_c = \sqrt{\mu_g}$

Say for example if $\mu_g = 1.6$ then $\mu_c = 1.2649$

Thus the refractive index of the coating material should be less than glass and should be greater than air. i.e. $\mu_g > \mu_c > \mu_a$. If, the incident light of wavelength λ is incident on the surface of the coating material, equal quantities of light are reflected from the reflecting surface. Therefore from the theory of interference of thin films, the path difference between these two rays is $2\mu_c t \cos r$.

Where t is the thickness of the coating given

For normal incidence $r = 0$; $\cos r = 1$

$$\therefore \text{Path difference} = 2\mu_c t \dots\dots\dots(4)$$

$$\text{For destructive interference, path difference} = \frac{\lambda}{2} \dots\dots\dots(5)$$

$$\text{Equating eq. (4) and eq. (5) we get } 2\mu_c t = \frac{\lambda}{2} \text{ or } t = \frac{\lambda}{4\mu_c}$$

In this case, the thickness of the film $t = \frac{\lambda}{4\mu_c}$ will be true for one particular wavelength. For this

purpose, the wavelength (λ) is generally chosen near the middle of the visible spectrum, where the eye is most sensitive. This is a very useful method to decrease the reflection to a minimum value. Such thin coatings of a liquid of proper thickness are known as **non - reflecting films**. These non - reflecting coatings are made by either.

- (i) Evaporating calcium or magnesium fluoride on the surface of glass in vacuum, or
- (ii) By chemical treatment with acids which leave a thin layer of silica on the surface of the lens

Application of Antireflection Coating

- (i) It is used in lens system of cameras
- (ii) It reduces the reflection and improves the transmission.
- (iii) It is used in optical instruments such as periscope, telescope etc.,

2.8. INTERFERENCE FILTERS:

We know white light consists of all the colours ranging from wavelength $4,000\text{\AA}$ to $7,000\text{\AA}$. Out of these colours if we need a particular colour say green colour of wavelength ($5461\text{\AA}$) as the output and the other to be filtered, then we can use interference filters. The interference filter consists of a thin film of dielectric medium (Quartz, air or MgF_2) of thickness ' t ', placed in between two reflecting films as shown in Fig. 2.13 (a & b)

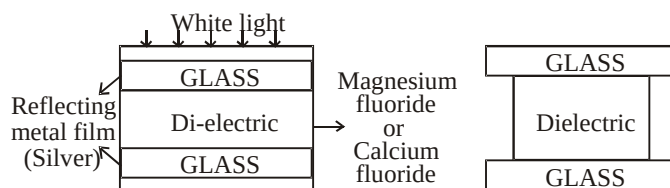


Fig. 2.13 (a)

Fig. 2.13 (b)

Fabry-Perot Interferometer Setup

The either end of the reflecting films is covered by the glass plates. Now, if white light is passed

through the filter, based on the thickness of the thin film of dielectric medium we get the required output colour. For example if we want green colour light from the visible region as the output we should make the thickness of the film as 560 nm, because of the following reason.

From the theory of interference due to transmitted light, for output (brightness) to come, the condition is $2\mu t \cos r = n\lambda$, for normal incidence $r=0$ $\cos 0=1$ and if $\mu=1$ for air then

$$\therefore 2t = n\lambda$$

or Wavelength of light emitted by the filter

$$\lambda = \frac{2t}{n} \dots\dots\dots (1)$$

Let us see the possibilities of the output wavelength from the filter.

In equation (1) 'n' can take integer (i.e) 1,2,3n

(i) If $t = 560 \text{ nm}$; $n = 1$, eq. (1) becomes $\lambda = \frac{2 \times 560 \times 10^{-9}}{1} = 11200 \text{ \AA}$

(Wavelength corresponds to infra red region)

(ii) If $t = 560 \text{ nm}$; $n = 2$ eq. (1) becomes

$$\lambda = \frac{2 \times 560 \times 10^{-9}}{2} = 5600 \text{ \AA} \approx 5461 \text{ \AA}$$

(This Wavelength corresponds to visible region (green))

(iii) If $t = 560 \text{ nm}$; $n = 3$, eq. (1) becomes

$$\lambda = \frac{2 \times 560 \times 10^{-9}}{3} = 3733 \text{ \AA} \text{ (This wavelength corresponds to ultraviolet region)}$$

(iv) If $t = 560 \text{ nm}$; $n = 4$, eq. ... (1)

$$\lambda = \frac{2 \times 560 \times 10^{-9}}{4} = 2800 \text{ \AA} \text{ (This wavelength also corresponds to UV region)}$$

Thus, out of these four conditions we can see that the filter can allow the green colour (5600Å) alone in the visible region and all the other colours in the visible region are absorbed (or) filtered. Hence the dielectric acts as an interference filter. These filters are used in spectroscopy to get a narrow wavelength.

2.9. DIFFRACTION :

Introduction

When a beam of light passes through a narrow aperture, it always spreads to some extent into the region which is not directly exposed to the incoming light waves. Such phenomenon of spreading of light when it passes through a narrow aperture or bending of light around the edges of an opaque body is known as **diffraction of light**. If the dimension of the aperture is very large compared to the wavelength of the light, the diffraction effect is not prominent. On the other hand, the diffraction is prominent if the dimension of the aperture is comparable with the wavelength of the light used.

Careful examination of light shows that light does not travel in a straight line when it is passing through an obstacle comparable to its wavelength. The diffraction phenomenon is also observed in sound waves, water waves, microwaves and so on. Fresnel showed that the amount of bending produced by an obstacle is directly proportional to the wavelength of light used therefore, in sound waves, the diffraction is more pronounced than in light waves. Diffraction is classified into two types namely

1. Fresnel diffraction, and
2. Fraunhofer diffraction

2.9.1. Fresnel Diffraction:

In Fresnel diffraction the source of light and the screen are separated by a finite distance. In this case, the wavefront is either spherical or cylindrical and diffraction is obtained without using any lens Fig. 2.14 (a).

Fraunhofer Diffraction:

In Fraunhofer diffraction the source of light and screen are placed at infinite distance. To bring the diffracted ray into focus, a convex lens is placed in between the slit and the screen Fig. 2.14 (b).

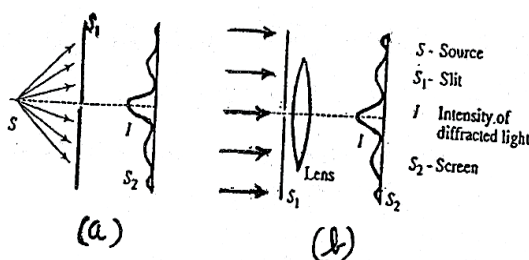


Fig. 2.14 (a)

Fig. 2.14 (b)

2.10. Difference between Interference and Diffraction

Interference	Diffraction
(i) Interference bands are of equal width	Diffraction fringes are not of the same width
(ii) All bright bands are of uniform intensity.	All bright bands are not of the same intensity
(iii) Points of minimum intensities are perfectly dark	Points of minimum intensities are not perfectly dark.
(iv) Interference takes place when two different coherent wave fronts originating from the same source superimpose each other	Diffraction pattern is the result of interaction of light coming from different parts of the same wave front.

2.11. DIFFRACTION GRATING

Diffraction grating is an arrangement consisting of a large number of parallel and equidistant slits of the same width (10^{-6}m) which have small spacing between them. Diffracted beams from each slit interfere with each other and produce a diffraction pattern consisting of maxima on the screen only at those points where the effects reinforce with one another.

Diffraction grating is constructed by drawing equidistant parallel lines on a transparent material

like glass plate by means of a diamond edge. The rulings made on the glass plate act as an opaque medium for the light, whereas the spacing between any two lines act as a transparent medium for the incident light. In a plane transmission grating nearly 1500 lines are drawn in one inch space.

2.12. THEORY OF DIFFRACTION GRATING

Let AB, CD, EF..... MN, be the transparent spaces of width 'a'. Let BC, DE, FG NO be the opaque spaces of width 'b'. The distance a+b is known as grating element. Consider a parallel beam of light which is made to incident on the grating. The transparent spaces AB, CD, EF MN acts as source for secondary wavelets as shown in Fig. 2.15. Consider the different wavelets proceeding towards the lens L, and making an angle of θ with the incident wavefront line AMN, is normal to the direction of propagation of the diffracted beam. Consider the wavelets starting from the points A and C. The path difference between them, on reaching the wavefront AMN, is CM.

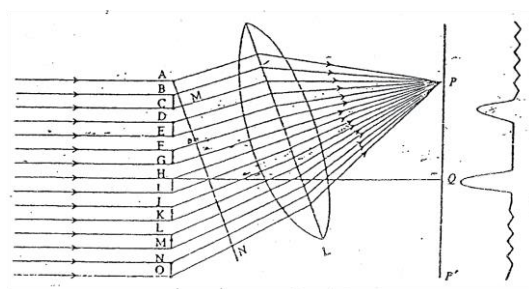


Fig. 2.15.

From the above figure 2.15

$$\begin{aligned} CM &= AC \sin \theta \\ &= (a+b) \sin \theta \end{aligned}$$

For constructive interference, the path difference is equated to integral multiple of λ

$$\text{i.e., } (a+b) \sin \theta = n\lambda \quad \text{..... (1)}$$

where n is an integer and it can take the value of 1,2,3,.....etc.,

For a grating, $\frac{1}{a+b}$ gives the number of lines drawn on the grating (N).

$$\text{Therefore eq (1) can be written as } \sin \theta = Nn\lambda \quad \text{..... (2)}$$

It is clear from the above discussion that light passes through the grating and, hence, is known as **transmission grating**. Diffraction is also observed when light is reflected from the surface of the grating. Such a grating is known as **reflection grating**.

Case (i)

For $n = 0$, $\sin \theta = 0$, we get maximum intensity at O. This represents the direct beam.

Case (ii)

For $n = 1$, $\sin \theta = N\lambda$ we get the first order diffraction pattern. When a monochromatic beam of light is used, there will be bright bands on either side of the direct beam. On the other hand, if a white beam of light is used, different coloured bands are produced on either side of the direct beam. This is called **grating spectrum**.

Case (iii)

For $n = 2$, second order diffraction pattern is produced on either side of the first order spectrum, When a beam of white light is used, coloured bands are obtained along either side of the first order spectrum.

Maximum order of spectrum

The maximum order of spectrum in a diffraction grating is obtained by substituting $\sin \theta = 1$ in eq (2).

Therefore eq (2) can be written as

$$1 = nN\lambda$$

$$n = \frac{1}{N\lambda}$$

For a wavelength of $\lambda = 6000\text{\AA}$ and $= 6 \times 10^5$ lines / m, we get,

$$n = \frac{1}{6000 \times 10^2 \times 6000 \times 10^{-10}}$$

(or) $n = 2.77$

Therefore, the maximum order that can be observed is 2.

2.13. RESOLVING POWER OF OPTICAL INSTRUMENTS

When the two objects are very near to each other or they are at very large distance from our eye, the eye may not be able to see them as separate. If we want to see them separate, optical instruments such as telescope, microscope etc. for close objects and prism and grating etc. for spectral lines are employed. Even if we assume that the instruments employed are completely free from all optical defects, the image of a point object or line is not simply a point or line but it is a diffraction pattern with a bright central maximum and other secondary maxima having minima in between of rapidly decreasing intensity. Thus an optical instrument is said to be able to resolve two point objects if the corresponding diffraction patterns are distinguishable from each other. **"The ability of the instrument to produce their separate patterns is known as resolving power"**

2.14. RAYLEIGH'S CRITERION OF RESOLUTION

"According to Rayleigh criterion, two point sources are resolvable by an optical instrument when the central maximum in the diffraction pattern of one falls over the first minimum in the diffraction pattern of the other and vice versa."

Similarly in case of spectral lines of two different wavelengths, the lines will be resolved when the central maximum due to one wavelength falls over the first minimum due to other and vice versa. In order to illustrate the criterion let us consider the resolution of two wavelengths λ_1 and λ_2 by a grating. The Fig. 2.16 (a) shows the intensity curves of the diffraction patterns of two wavelengths.

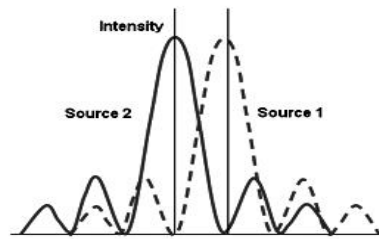


Fig. 2.16 (a)

The difference in wavelength is such that their principal maxima are separately visible. There is a distinct point of zero intensity in between the two. Hence the two wavelengths are well resolved.

Now consider the case when the difference in wavelengths is smaller and such that the central maximum of wavelengths coincides with the first minimum of the other as shown in the below Fig. 2.16 (b)

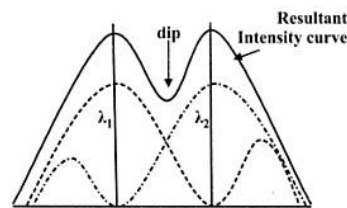


Fig. 2.16 (b)

The resultant intensity curve is shown by thick curve. The curve shows a distinct dip in the middle of two central maxima i.e. there is a noticeable decrease in intensity between the two central maxima indicating the presence of two different wavelengths. Thus the two wavelengths can distinguished from one another and according to Rayleigh they are said to just resolved. Again consider the case when the difference in wavelengths is so small that the central maxima corresponding to two wavelengths come still closer as shown in the below Fig. 2.17 (c)

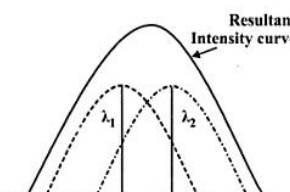


Fig.2.17(c)

The resultant intensity curve in this case is quite smooth without any dip thus giving the impression as if there is only one wavelength source although somewhat bigger and stronger. Hence the two wavelengths are not resolved. Thus the two spectral lines can be resolved only upto a certain limit expressed by Rayleigh criterion.

2.15. Difference between Dispersive power and Resolving power

Dispersive power	Resolving power
1. The dispersive power of a grating gives an idea of angular separation between two spectral, lines produced by grating	Resolving power tells the limit of just resolution of two close objects.
2. Dispersive power is measured by $\frac{d\theta}{d\lambda}$ where $d\theta$ - the angular separation between two lines	Resolving power is measured by $\frac{\lambda}{d\lambda}$ and given by $\frac{\lambda}{d\lambda} = nN$ Where $N \rightarrow$ No. of lines on the grating surface.
3. When N (Number of lines on grating surface) is increased the dispersive power remains unchanged.	When N (Number of lines on grating surface) is increased the resolving power is increased.
4. If $(a+b)$ (the grating element) is decreased the dispersive power is increased	If $(a+b)$ (the grating element) is decreased the resolving power remains unchanged

2.16. RESOLVING POWER OF A PRISM

The term resolving power is applied to spectrographic devices using a prism or a grating. Resolving power signifies the ability of the instrument to form separate spectral images of two neighboring wavelengths λ and $\lambda + d\lambda$ in the wavelength region λ . In the Fig.2.18, S is a source of light, L_1 is a collimating lens and L_2 is the telescope objective.

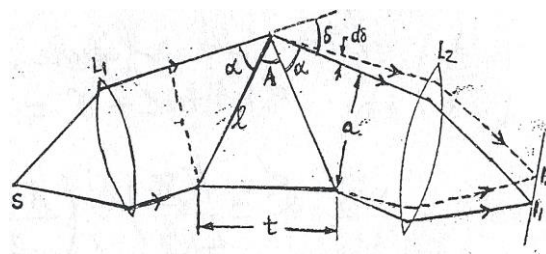


Fig.2.18

As the two wavelength λ and $\lambda + d\lambda$ are very close, if the prism is set in the minimum deviation position it would holds good for both the wavelengths the final image I_1 corresponds to the principal maximum for wavelength λ and image I_2 corresponds to the principal maximum for wavelength $\lambda + d\lambda$. I_1 and I_2 are formed at the focal plane of the telescope objective lens L_2 . The face of the prism limits the incident beam to a rectangular section of width 'a'. Hence, the Rayleigh criterion can be applied in the case of a rectangular aperture.

In the case of diffraction at a rectangular aperture, the position of I_2 will correspond to the first minimum of the image I_1 for wavelength λ provided. $a \cdot d\delta = \lambda$

$$d\delta = \frac{\lambda}{a} \dots \dots \dots (1)$$

From the Figure.2.18.

$$\alpha + A + \alpha + \delta = \pi$$

$$\alpha = \left[\left(\frac{\pi}{2} \right) \right] - \left(\frac{A + \delta}{2} \right)$$

$$\sin \alpha = \sin \left[\frac{\pi}{2} - \left(\frac{A + \delta}{2} \right) \right]$$

$$\sin \alpha = \cos \left(\frac{A + \delta}{2} \right)$$

$$\text{But } \sin \alpha = \frac{a}{l}$$

$$\therefore \cos \left(\frac{A + \delta}{2} \right) = \frac{a}{l} \dots\dots\dots (2)$$

$$\text{Also } \sin \frac{A}{2} = \frac{t}{2l} \dots\dots\dots (3)$$

In the case of a prism

$$\mu = \frac{\sin \left(\frac{A + \delta}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

$$\sin \left(\frac{A + \delta}{2} \right) = \mu \sin \frac{A}{2} \dots\dots\dots (4)$$

Here μ and δ are dependent on wavelength of light λ

Differentiating equation (4) with respect to λ

$$\frac{1}{2} \left(\cos \left[\frac{A + \delta}{2} \right] \right) \frac{d\delta}{d\lambda} = \frac{d\mu}{d\lambda} \left[\sin \left(\frac{A}{2} \right) \right] \dots\dots\dots (5)$$

Substituting the values from equations (2) and (3) in eq (5)

$$\frac{1}{2} \left(\frac{a}{l} \right) \frac{d\delta}{d\lambda} = \frac{d\mu}{d\lambda} \left(\frac{t}{2l} \right)$$

$$a. \frac{d\delta}{d\lambda} = t. \frac{d\mu}{d\lambda} \dots\dots\dots (6)$$

Substituting the values of $d\delta$ from equation (1)

$$\frac{\lambda}{d\lambda} = t. \frac{d\mu}{d\lambda} \dots\dots\dots (7)$$

The expression $\frac{\lambda}{d\lambda}$ measures the resolving power of the prism. It is defined as the ratio of the

wavelength of a spectral line to the difference in wavelength between this line and a neighboring line such that these lines are just resolved according to **Rayleigh's Criterion**.

∴ Resolving power of a prism is given by $t \frac{d\mu}{d\lambda}$

Thus, the resolving power of a prism is directly proportional to

- (i) Length of the base of the prism and
- (ii) Rate of change of refractive index with respect to wavelength for that particular material.

The expression for resolving power given above is applicable only to spectral lines of equal intensity. If two spectral lines are of different intensities, then the value of $d\lambda$ i.e., the difference in wavelength between the two lines must be higher so that the two lines appear as separate ones.

2.17. RESOLVING POWER OF A PLANE DIFFRACTION GRATING:

The resolving power of a grating is defined as the ratio of the wavelength of any spectral line to the difference in wavelength between this line and a neighboring line such that the two lines appear to be just resolved. Thus the resolving power of a grating is equal to $\frac{\lambda}{d\lambda}$

In the Fig. 2.19, XY is the grating surface and MN is the field of view of the telescope. P_1 is the n^{th} primary maximum of a spectral line of wavelength λ at an angle of diffraction θ_n , P_2 is the n^{th} primary maximum of a spectral line of wavelength $\lambda + d\lambda$ at an angle of diffraction $\theta_n + d\theta$

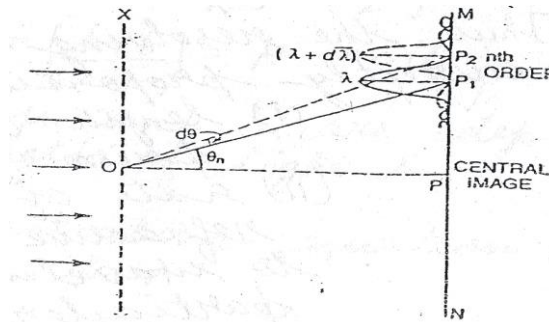


Fig. 2.19

Let P_1 and P_2 are the spectral lines in the n^{th} order. These two spectral lines according to Rayleigh, will appear resolved if the position of P_2 also corresponds to the first minimum of P_1 .

The direction of the n^{th} primary maximum for a wavelength λ is given by

$$(a+b) \sin \theta_n = n\lambda \dots\dots\dots(1)$$

The direction of the n^{th} primary maximum for a wavelength $(\lambda + d\lambda)$ is given by

$$(a+b) \sin (\theta_n + d\theta) = n(\lambda + d\lambda) \dots\dots\dots(2)$$

The two lines will appear just resolved if the angle of diffraction $(\theta_n + d\theta)$ also corresponds to the direction of the first secondary minimum after the n^{th} primary maximum at P_1 (corresponds to wavelength λ). This is possible if the extra path difference introduced is $\frac{\lambda}{N}$ where N is the total number of lines on the grating surface.

$$\therefore (a + b) \sin(\theta_n + d\theta) = n\lambda + \frac{\lambda}{N} \dots\dots\dots(3)$$

Equating, the RHS of equation (2) and

$$n(\lambda + d\lambda) = n\lambda + \frac{\lambda}{N}$$

$$nd\lambda = \frac{\lambda}{N}$$

$$nd\lambda = \frac{\lambda}{N} \dots\dots\dots(4)$$

The quantity $\frac{\lambda}{d\lambda} = nN$ measures the resolving power of a grating. Thus, the resolving power is

independent of the grating constant. The resolving power is directly proportional to

- (i) the order of the spectrum and
- (ii) the total number of lines on the grating surface.

For a given grating, the spacing between the spectral lines is double in the second order than that in the first order.

2.18. DISPERSIVE POWER OF A GRATING

Dispersive power of a grating is defined as the ratio of the difference in the angle of diffraction of any two neighboring spectral lines to the difference in wavelength between the two spectral lines. It can also be defined as the difference in the angle of diffraction per unit change in wavelength. The diffraction of the n^{th} order principal maximum for a wavelength λ is give by the equation,

$$(a + b) \sin \theta = n\lambda \text{ -----}(1)$$

Differentiating this equation with respect to θ and λ ($a + b$) is constant and n is constant in a given order

$$(a + b) \cos \theta d\theta = nd\lambda$$

$$\frac{d\theta}{d\lambda} = \frac{n}{(a + b) \cos \theta}$$

$$\frac{d\theta}{d\lambda} = \frac{nN'}{\cos \theta} \text{ ----- } (2)$$

In the equation (2) $\frac{d\theta}{d\lambda}$ is the dispersive power, n is the order of the spectrum, N' is the number of lines per cm of the grating surface and θ is the angle of diffraction for the n^{th} order principal maximum of wavelength λ . From equation (2), it is clear that the dispersive power of the grating is

- (i) Directly proportional to the order of the spectrum
- (ii) Directly proportional to the number of lines per cm and
- (iii) Inversely proportional to $\cos \theta$.

Thus, the angular spacing of any two spectral lines is double in the second order spectrum in comparison to the first order. Secondly, the angular dispersion of the lines is more with a grating having larger number of lines per cm. Thirdly, the angular dispersion is minimum when $\theta = 0$. If the

value of θ is not large the value of $\cos \theta$ can be taken as unity approximately and the influence of the factor $\cos \theta$ in the equation (2) can be neglected. Neglecting the influence of $\cos \theta$, it is clear that the angular dispersion of any two spectral lines (in a particular order) is directly proportional to the difference in wavelength between the two spectral lines. The spectrum of this type is called a **normal spectrum**.

2.19. DIFFERENCES BETWEEN GRATING AND PRISM SPECTRA

Prism Sepctra	Grating Spectra
1. Prism produces only one spectrum	A grating gives a number of spectra on either side of the direct image.
2. This spectrum is pure one	This is impure at the higher orders due to overlapping of the spectral lines
3. This will give a bright spectrum. Since all light energy is concentrated on one spectrum	Most of the energy goes into zero order where no spectrum is formed the rest of the energy is distributed among the spectra of various order.

POLARISATION

2.20. INTRODUCTION

Explanation of interference and diffraction phenomenon leads to the wave nature of light. But it does not tell about the character of the light wave whether it is longitudinal or transverse. Transverse nature of light was postulated by Thomas Young in 1817 to explain the absence of interference between light waves vibrating in mutually perpendicular planes.

The natural light is unpolarised which can be transformed into different types of polarization using optical devices. Light is an electromagnetic wave consisting of electric and magnetic field vibrating perpendicular to each other and to the direction of propagation. Since electric vector is more effective therefore most of the time we refer to the plane of vibration constituted by electric vector and direction of propagation in describing the polarization. Polarized light has many applications in the engineering industry.

Polarisation of light phenomenon confirms about the transverse nature of light. When light is passed through a pair of crystal plates (P_1 and P_2) called “**Tourmaline**” (occurring in nature) with their planes at right angles to the direction of light, the intensity is maximum as shown in Fig 2.20(a). However, when the plane of P_2 is rotated through 90° the intensity is minimum. This is shown in the Fig.2.20(b). So, by this way we have restricted the vibrations in one particular direction only so that in other non-parallel orientation of the crystal, light is not allowed pass through. This shows that ordinary light is transverse in nature where the vibrations are in perpendicular direction to the direction of propagation of the wave.

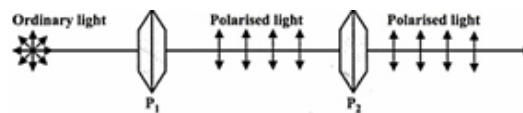


Fig. 2.20(a)

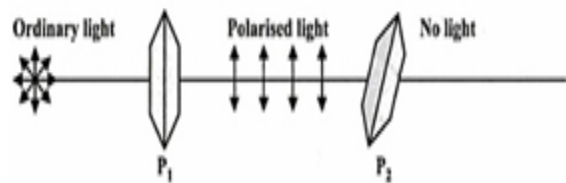


Fig. 2.20(b)

The phenomenon of confinement of electric vector vibration in certain direction about the direction of propagation of light is called as polarization of light. Thus this phenomenon may be defined as asymmetrical light vector vibration about the direction of propagation of light. It means light having the property of one sidedness is called **polarized light**.

2.21 FUNDAMENTALS OF POLARISATION

UNPOLARISED LIGHT

If the vibration of a light source is in all possible directions with respect to direction of propagation that light is defined as **unpolarised light**.

POLARISED LIGHT

If the vibration of light is perpendicular to the direction of propagation than that light is defined as **polarised light**.

POLARISER

A device which converts an unpolarised light into a plane polarised light is called a **polariser**.

ANALYSER :

A device which checks whether the light is plane polarised or not is called an **analyser**. e.g. nicol Prism

NICOL PRISM

It is an optical device made up of calcite to produce and analyse plane polarised light.

DOUBLE REFRACTION

When light is passed through a calcite crystal, the single incident beam splits into two reflected rays.

- (i) Ordinary ray (O-ray)
- (ii) Extra - ordinary ray (E-ray)

2.22. HUYGEN'S THEORY OF DOUBLE REFRACTION

According to his theory when any wavefronts strikes a doubly refracting crystal, every point of the crystal becomes a source of two wavefronts.

a) Ordinary wavefront corresponding to ordinary rays. Since ordinary rays have same velocity in all directions, the secondary wavefront is spherical.

b) Extra - ordinary wavefront corresponding to extra - ordinary rays. Since extra - ordinary rays have different velocities in different directions, the extra ordinary wavefront is ellipsoid of revolution, with optic axis as the axis of revolution.

The sphere and ellipsoid touch each other at points which lie on the optic axis of the crystal, because the velocity of ordinary and extra - ordinary ray is same along the optic axis. In certain crystals (like calcite and tourmaline) called the **negative crystals**, the ellipsoid lies outside the sphere as shown in Fig. 2.21 (a)

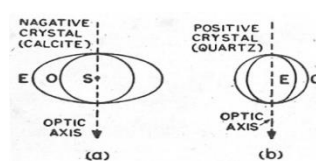
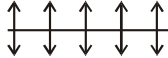


Fig. 2.21 (a) & b

This shows that in negative crystals, the extra ordinary wavefront travels faster than ordinary wavefront except along optic axis. In certain crystals (like those of ice and quartz) called the positive crystals, the sphere lies outside the ellipsoid as shown in Fig. 2.21 (b). This shows that velocity of ordinary wavefront is greater than extra - ordinary wavefront (in positive crystals) except along optic axis. Now we shall consider the phenomenon of double refraction in the following different cases.

2.23. Differences between O - ray and E – ray

Ordinary Ray	Extra - Ordinary Ray
(i) It obeys the laws of refraction	It does not obey the laws of refraction
(ii) It travels with a low velocity	It travels with a higher velocity than ordinary ray
(iii) Ordinary ray is stationery	Extra - ordinary ray revolves around the ordinary ray
(iv) It is denoted by 	It is denoted by 

2.24. WAVE PLATE

We know refracting crystals like calcite, quartz etc. will produce double refraction. Normally we have two types of crystals viz.

- Uniaxial crystals, which has only one optic axis
- Bi-axial crystals, which has two optic axis

The uniaxial crystals are further classified into two types

- Positive uniaxial crystal, for which $\mu_e > \mu_o$, example : Quartz.
- Negative uniaxial crystal for which $\mu_o > \mu_e$ example : Calcite

Definition:

Waveplate is nothing but a transparent plate made of uniaxial crystals like quartz (or) calcite, cut parallel to the optic axis. When light falls on a wave plate at 90° to the optic axis it splits up into two rays

- Ordinary ray and 2. Extra ordinary ray

Thus a path difference and hence a phase difference is introduced between these two rays. Based on the path difference or phase difference we have two types of wave plates

- Quarter wave plate and
- Half wave plate.

2.24.1. QUARTER WAVE PLATE :

Quarter wave plate is doubly refracting material made of uniaxial crystal whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path difference of $\frac{\lambda}{4}$ (or) a phase difference of $\frac{\pi}{2}$ between the ordinary and extra ordinary rays, emerging from it.

Thickness of Positive uniaxial crystal

In the case of positive uniaxial crystal of thickness 't' the velocity of O-ray is greater than the velocity of E-ray. Therefore, the refractive index of O-ray will be lesser than the E-ray.

$$\therefore \mu_e > \mu_o$$

$$\therefore \text{Path difference} = t(\mu_e - \mu_o) = \frac{\lambda}{4}$$

$$\text{(or) Thickness } t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

Thickness of negative uniaxial crystal

In the case of negative uniaxial crystal of thickness 't', the velocity of E-ray is greater than O-ray. Therefore, the refractive index of E-ray, is lesser than the O-ray ($\mu_o > \mu_e$).

$$\therefore \text{Path difference} = t(\mu_o - \mu_e) = \frac{\lambda}{4}$$

$$\text{(or) Thickness } t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

Applications :

(i) When the vibrations of the polarised light, inclined at an angle of 45° to the optics is made to incident on a quarter wave plates the emergent beam becomes circularly polarised light. Therefore quarter wave plate is used to convert plane polarised light to circularly polarised light and vice versa.

(ii) When the vibrations of the polarised light inclined at an angle not equal to 45° (i.e. other than 45°) to the optic axis is made to incident on a quarter wave plate, the emergent beam becomes elliptically polarised light. Thus it is used to convert plane polarised light to elliptically polarised light and vice versa.

Thus a quarter wave plate can be used to convert.

(i) Plane polarised light $\rightleftharpoons$ Elliptically polarised light

(i) Plane polarised light $\rightleftharpoons$ Circularly Polarised light

2.24.2. Half Wave Plates

Definitions :

Half wave plate is a doubly refracting material made of uniaxial crystal like quartz (or) calcite, whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path different of $\frac{\lambda}{2}$ or phase difference π between the ordinary - ray and the extra ordinary ray emerging from it

$$\text{Thickness of positive uniaxial crystal is } t = \frac{\lambda}{2(\mu_e - \mu_o)}$$

Thickness of negative uniaxial crystal is $t = \frac{\lambda}{2(\mu_o - \mu_e)}$

APPLICATION :

When a plane - polarised light is incident on a half - wave plate such that it makes an angle of 45° with the optic axis, then the emergent beam becomes a plane - polarised light with 90° rotation. Thus the half wave plate is capable of rotating the angle of the plane polarised light by 90°

2.25. Optical Activity

When a polarizer and an analyser are crossed, no light emerges out of the analyser as shown in Fig. 2.30

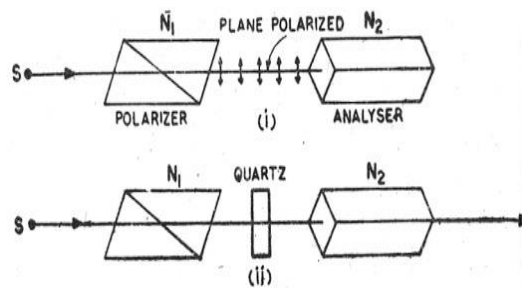


Fig. 2.30

When a quartz plate cut with its faces parallel to the optic axis is introduced between N_1 and N_2 such that light falls normally upon the quartz plate, the light emerges out of N_2 .

The quartz plate turns the plane of vibration. The plane polarised light enters the quartz plate and its plane of vibration is gradually rotated as shown in Fig. 2.31

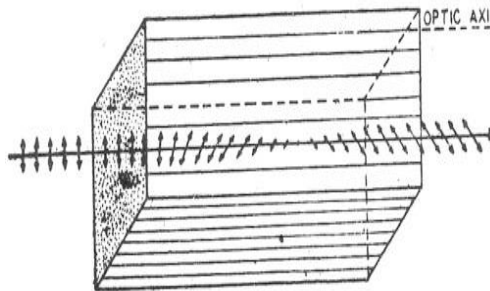


Fig. 2.31

The amount of rotation through which the plane of vibration is turned depends upon the thickness of the quartz plate and the wavelength of light. The action of turning the plane of vibration occurs inside the body of the plate and not on its surface. This phenomenon or the property of rotating the plane of vibration by certain crystals or substances is known as **optical activity** and the substance is known as an **optically active substance**. It has been found that calcite does not produce any change in the plane of vibration of the plane polarised light. Therefore, it is not optically active.

Substances like sugar crystals, turpentine, sodium chlorate and cinnabar are optically active. Some of the substances rotate the plane of vibration to the right and they are called **dextro - rotatory**

or right handed. Right handed rotation means that when the observer is looking towards light travelling towards him, the plane of vibration is rotated in a clockwise direction. The substances that rotate the plane of vibration to left (anti - clockwise) from the point of view of the observer are known as **laevo - rotatory or left - handed.**

It has been found that some quartz crystals are dextro - rotatory while others are leavo - rotatory. One is the mirror image of the other in their orientation the rotation of the plane of vibration in a solution depends upon the concentration of the optically active substance in the solution. This helps in finding the amount of cane sugar present in a sample of sugar solution.

2.26. SPECIFIC ROTATION

Liquids containing an optically active substance e.g. sugar solution, camphor in alcohol etc., rotate the plane of the linearly polarised light. The angle through which the plane polarised light is rotated depends upon.

1. The thickness of the medium
2. Concentration of the solution or density of the active substance in the solvent.
3. Wavelength of light

The specific rotation is defined as the rotation produced by a decimeter (10 cm) long column of the liquid containing 1 gram of the active substance in one cc of the solution therefore

$$S = \frac{10\theta}{LC} \text{ deg/dm}$$

Where S represents the specific rotation at temperature $t^{\circ}\text{C}$ for a Wavelength, θ is the angle of rotation, L is the length of the solution in cm through which the plane polarised light passes and C is the concentration of the active substance in gm /cc in the solution. The angle through which the plane of polarisation is rotated by the optically active substance is determined with the help of Polarimeter. When this instrument is used to determine the quantity of sugar in a solution it is known as a "**Saccharimeter**".

2.27. LAURENT'S HALF SHADE POLARIMETER

Polarimeters are the instruments designed to measure the angle of rotation produced by a substance. When they are calibrated to read directly the percentage of cane sugar in a solution, they are named as Saccharimeters Polarimeters can be used to find the specific rotation of sugar solution or if the specific rotation is known, they can be used to find its concentration.

CONSTRUCTION

The essential parts of a Laurent's half - shade Polarimeter as shown in Fig. 2.32

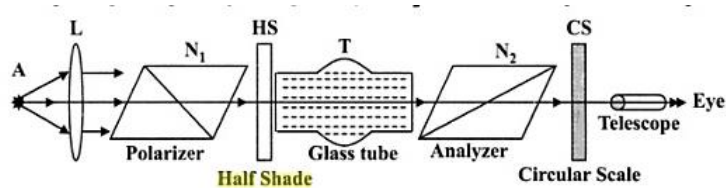


Fig. 2.32

'S' is a monochromatic source of light and L is a convex lens which renders the incident light into a parallel beam. N_1 and N_2 are two nicol prisms. N_1 acts as polariser while N_2 act as analyser. N_2 is capable of rotation about a common axis of N_1 and N_2 . The rotation of analyser (N_2) can be read on a graduated circular scale (C S.). The verniers are also provided to read the fraction of a degree. Light after passing through polariser becomes plane polarised with its vibrations in the principal plane of the Nicol N_1 . The plane polarised light now passes through a half shade device (H.S.) and then through a tube T containing the optically active substance (say sugar solution.)

Usually T is a hollow glass tube having a large diameter in the middle so that no air bubble may be in the path of light when filled with a liquid. This tube is closed at the ends by cover - slips and metal covers. The emergent light on passing through analyser N_2 is viewed through a telescope T. The telescope is focussed on the half - shade.

ACTION OF HALF - SHADE

When an optical active substance is placed in between two crossed Nicols the field of view is not dark. In order to make the field of view dark, the analyser is rotated. It is observed that when the analyser is rotated the field of view is not dark, for a considerable region. Hence the measurement of optical rotation is not accurate. To avoid this difficulty a half shade device is used. The Laurent's half - shade plate consists of a semi circular half - wave plate ABC of quartz (cut parallel to optic axis) so that it introduces a phase change of π between extra - ordinary and ordinary rays passing through it and a semicircular glass plate ADC as shown in Fig. 2.33.

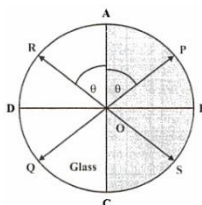


Fig. 2.33.

The two plates are cemented along AC. The thickness of glass plate is such that it absorbs the same amount of light as the quartz plate. Let the plane of vibration of the plane polarized light incident normally on the half shade plate is along PQ. PQ makes an angle θ with AC as in Fig. 2.33. The vibrations emerge from glass plate along PQ. Inside quartz plate light is divided into two components,

one, O-ray along XX', and the other, E-ray, parallel to optic axis, YY' as shown in Fig.2.34. The two components travel along same direction but with different speeds. As quartz is a positive uniaxial doubly refracting crystal.

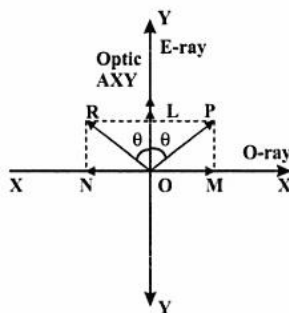


Fig. 2.34

E-ray component moves with higher velocity than O-ray component and on emergence a phase difference of π introduced between them. Due to this phase difference the direction of O components reversed. If initial direction is OM, then, it finally becomes ON. So, the resultant of E component OL and ON will be OR which makes an angle θ with Y-axis.

The vibration of beam of emergent light out of quartz will be along RS. If the principal plane of N_2 is along PQ, the light pass unobstructed through glass plate ADC, but partly obstructed by ABC, so the glass half will appear brighter than the quartz half as shown in Fig. 2.35 (a). Other way, if the principal plane of N_2 is along RS, reverse situation occurs, so then quartz half will appear brighter than the glass half as shown in Fig. 2.35(b).

However, if the principal plane of N_2 is parallel to AC (Y-axis), it is equally inclined both for O and E rays. The field of view becomes equally illuminated as shown in Fig. 2.35(c). Thus, half shade serves the purpose of dividing the field of view in two halves. When there is a rotation of N_2 from the position of equal brightness or illumination a marked change in the intensity of two halves is observed.

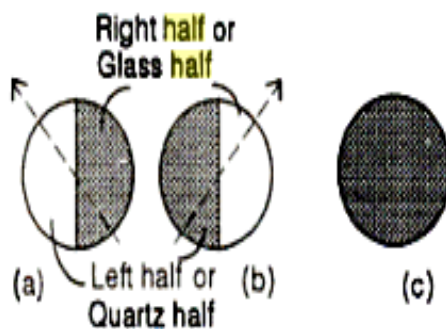


Fig. 2.35 (a) 2.35 (b) 2.35 (c)

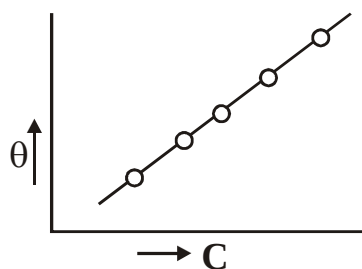
Determination of specific Rotation of sugar solution by Laurents Half shade polarimeter

We know that the specific rotation S is given by

$$S = \frac{10\theta}{LC} \text{ deg/dm cc}$$

Where θ is the angle of rotation in degrees, l is the length of the solution in decimeters and C is the concentration of solution in gm per cc. Hence to determine the specific rotation of a substance solution of known concentration is prepared. The length of the solution is measured directly. The value of θ is determined as follows. First of all the experimental tube is filled with water and placed in its position. The telescope is focused on the half shade and the analyser is rotated till equally bright position is obtained. The readings of two verniers are noted on the circular scale. Now the tube is filled with the optically active solution already prepared and placed it in its position. The field of view in this position is equally bright. The analyser is rotated and is brought to a position so that the whole field of view is equally bright. The new positions of the two verniers are again noted on the circular scale. The difference in the two readings of the same verniers gives the angle of rotation produced by the solution. The experiment is repeated with different concentrations of the solution.

A graph is then plotted between concentration and angle of rotation as shown below. This comes out a straight line.



From the graph the ratio θ/C is determined. Substituting these values, the specific rotation is calculated by using the above formula.

Pondicherry University 2 marks Question & Answer

1. What is an air wedge? (Jan 2010, April 2010)

A wedge shaped air film is enclosed in between two glass plates is known as airwedge.

2. What is quarter wave plate? (Jan 2014, May 2017)

Quarter wave plate is doubly refracting material made of uniaxial crystal. Whose refracting faces are cut parallel to the optic axis. The thickness of this plate is in such a way that it produces a path difference of $\lambda/4$ or a phase difference of $\pi/2$ between the ordinary and extra ordinary rays emerging from it.

3. What is an anti reflection coating? (Jan 2011, 2016, May 2012)

A transparent material whose refractive index value lies between air and glass is coated on the surface the glass in order to prevent or to reduce reflection is called as anti reflection coating.

4. Explain the term Resolving power of an optical instrument. (Jan 2011)

The ability of the instrument to show the separation between the spectral lines of two wavelengths, which are very near to each other.

5. What is a circularly polarized light? (Jan 2010)

If light is composed of two plane waves of equal amplitude by differing in phase by 90° , then the light is said to be circularly polarized

6. What are uniaxial crystals? (May 2011)

A Uniaxial crystal is the one which has only one optic axis. A Bi-axial crystal is the crystal which has two optic axis.

7. What is half wave plate? (April, 2010, May 2011)

Half wave plate is a doubly refracting material made of uniaxial crystal whose refracting faces are cut parallel to the optics axis. The thickness of this plate is in such a way that it produces a path difference of $\lambda/2$ or a phase difference of π between ordinary ray and extraordinary ray emerging from it. The thickness of the half wave plate is to produce an optical path difference of $\lambda/2$ and is given by $t = \lambda / 2 (\mu_o - \mu_e)$

8. Explain the term specific rotation? (Jan 2013)

The specific rotation is defined as the rotation produced by a decimetre (10cm) long column of the liquid containing 1 gram of the active substance in one cc of the solution. It is given by $S = 100 / LC$ Where S – Specific rotation θ – angle of rotation L – length of the solution in cm and C is the concentration of the solution,

9. State the principle of Superposition of light.

(May 2012, April/May 2011, May 2017)

The principle states that resultant displacement of a particle of the medium acted upon by two or more waves simultaneously is the algebraic sum of the displacements of the same particle due to individual waves, in the absence of the others.

10. Define resolving power of a prism? (May 2013)

The resolving power of a prism is defined as its capacity to form separate spectral lines of two wavelengths which are very near to each other. It is given by $\frac{\lambda}{d\lambda} = t \cdot \frac{d\mu}{d\lambda}$

11. What do you understand by double refraction? (May 2013, Jan 2014)

When a light is passed through a calcite crystal, the single incident beam splits into two reflected rays (i) Ordinary ray (O-ray) (ii) Extra-ordinary ray (E-ray)

12. Distinguish between interference and diffraction? (Jan 2013, Jan 2014)

Sl.No	Interference	Diffraction
1	Interference takes place when two different coherent wave-fronts originating from the same source superimpose each other.	Diffraction pattern is the result of interaction of light coming from different parts of the same wave-front.
2	Interference bands are of equal width	Interference bands are not of equal width
3	All bright bands are of uniform intensity	All bright bands are not of the same intensity
4	Points of minimum intensities are perfectly dark	Points of minimum intensities are not perfectly dark

13. In a Michelson's interferometer 200 fringes cross the field of view when the movable mirror is displaced through 0.0589mm. Calculate the wavelength of monochromatic light used. (Jan 2011)

Given data:

$$d = 0.0589 \times 10^{-3} \text{m}$$

$$n = 200$$

Solution:

$$\text{Formula: Wavelength } \lambda = \frac{2d}{n}$$

$$= \frac{2 \times 0.0589 \times 10^{-3}}{200}$$

$$= 5.89 \times 10^{-7} \text{m}$$

$$= 589 \times 10^{-8} \text{m}$$

$$= 5890 \text{ \AA}$$

$$\text{Wavelength of light used} = 5890 \text{ \AA}$$

14. In an air wedge experiment, if the air gap between the wedge forming plates is filled with water what will happen to the fringe-width? (Jan 2014)

The fringe-width decreases because the fringe width β is inversely proportional to the refractive index which is given as $\beta = \lambda/2\mu\theta$.

15. Mention the applications of quarter wave plate (Jan 2016)

Applications :

(i) When the vibrations of the polarised light, inclined at an angle of 45° to the optics is made to incident on a quarter wave plates the emergent beam becomes circularly polarised light. Therefore quarter wave plate is used to convert plane polarised light to circularly polarised light and vice versa.

(ii) When the vibrations of the polarised light inclined at an angle not equal to 45° (i.e. other than 45°) to the optic axis is made to incident on a quarter wave plate, the emergent beam becomes elliptically polarised light. Thus it is used to convert plane polarised light to elliptically polarised light and vice versa.

Thus a quarter wave plate can be used to convert.

(i) Plane polarised light $\rightleftharpoons$ Elliptically polarised light

(i) Plane polarised light $\rightleftharpoons$ Circularly Polarised light

16. Define Resolving power of a grating.(May 2017)

The resolving power of a diffraction grating is defined as the capacity to form separate diffraction maxima of two wavelengths which are very close to each other. It is given by

$$\lambda/d\lambda = n.N$$

Additional 2 marks Question & Answer

1. Define Interference of light.

When two light waves superimpose, then the resultant amplitude or intensity in the region of superposition is different than the amplitude of individual waves. This modification in the distribution of intensity in the region of superposition is called interference.

2. What are the conditions to be satisfied for interference of light?

- i. The two sources should be coherent
- ii. The two sources must emit continuous waves of the same wavelength and time period.
- iii. The separation between the two sources ($2d$) should be small
- iv. The distance (D) between the two sources and screen should be large
- v. The sources must be narrow
- vi. The sources should be monochromatic
- vii. The amplitude of the interfering waves should be equal or nearly equal.

3. What is meant by fringe width?

The distance between any two consecutive bright and dark fringes is known as fringe width β . It is constant for both bright and dark fringe. It is given by $\beta = \lambda D/d$

4. What are the uses of Air Wedge?

Air Wedge is used to

- i. Find the thickness of very thin materials like hair, paper
- ii. To check the flatness of a given surface

5. Give the principle and applications of antireflection coating.

Antireflection or absorption coating, reduces the reflection. In other way we can say that it absorbs the incoming light, so that light is not wasted due to reflections. They are used in camera lens, telescope etc.,

6. Explain why colours are seen over a thin film of petrol on roads.

When white light, falls on a thin film of petrol on roads, interference takes place between the reflected light from the top and bottom layer of the petrol film and thus it appear coloured.

7. What are interference filters? Give its use.

Interference filters are made up of dielectric media like quartz or magnesium fluoride enclosed between reflecting films and covered by glass. They filter the incoming light and gives the required output light. They are used in spectroscopy to get a narrow wavelength.

8. What is meant by interferometry?

Instruments based on the principle of interference of light are known as interferometers and this branch of science is called interferometry.

9. What is the basic principle of Michelson's interferometer?

The amplitude of light beam from a source is divided into two parts of equal intensities by partial reflection and transmission. These beams then resent in two directions at right angles and are brought together after they suffer reflection form plane mirrors to produce interference fringes.

10. What is the role of compensating plate in Michelson's Interferometer?

In Michelson interferometer the ray which suffers reflection from mirror M_1 is passed through the glass plate G_1 twice whereas the ray reflected from mirror M_2 does not even passes once through the glass plate G_1 . so to compensate the path a second glass plate G_2 is introduced between the glass plate G_1 and the mirror M_2 .

11. What are the uses of Michelson interferometer?

It is used to determine

- (i) Wavelength of given monochromatic source
- (ii) The difference between the two neighboring wavelength or resolution of spectral lines.
- (iii) To find refractive index and thickness of various thin transparent materials
- (iv) For the measurement of the standard meter in terms of the wavelength of light.

12. What are the types of fringes in Michelson interferometer experiment? Mention the factor for the different types of fringes?

Types of fringes in Michelson Interferometer

- (i) Circular fringes
- (ii) Straight fringes
- (iii) White fringes

13. Define Diffraction.

The phenomenon of spreading of light when it passes through a narrow aperture or bending of light around the edges of an opaque obstacle is known as diffraction of light.

14. What are the two types of diffraction?

The two types of diffraction are

- (i) Fresnel diffraction and
- (ii) Fraunhofer diffraction

15. Define Fresnel Diffraction.

In this diffraction the source of light and the screen are separated by a finite distance. In this case the wave front is either spherical or cylindrical and the diffraction pattern is obtained without using any lens.

16. Define Fraunhofer Diffraction.

In this diffraction the source of light and screen are placed at infinite distance. To bring the diffracted ray into focus, a convex lens is placed in between the slit and the screen.

16. What is diffraction grating? (or) What is a plane transmission grating?

Diffraction grating or plane transmission grating is an arrangement consisting of a large number of parallel and equidistant slits of the same width (10^{-6}m) on a transparent material such as glass, with a diamond pointer. The ruled lines are opaque to light while the space between any two lines is transparent to light and act as grating. This is known as plane transmission grating.

18. What is Rayleigh's criterion of resolution.

According to Rayleigh criterion, two point sources are resolvable by an optical instrument when the central maximum in the diffraction pattern of one falls over the first minimum in the diffraction pattern of the other and vice versa.

19. Define Dispersive power of a grating?

The dispersive power of grating is defined as the ratio of variation of angle of diffraction with wavelength. It is given by

$$d\theta/d\lambda = n / (e+d) \cos \theta$$

20. What is meant by polarization of light?

The phenomenon of acquiring the property of one-sidedness in light is called polarization and the light is called polarized light.

21. What is an Unpolarised light.

If the vibration of a light source is in all possible directions with respect to direction of propagation is called as unpolarised light.

22. What is a plane polarized light?

When the vibration of the light are confined along a single direction of propagation the light is known as plane polarized.

23. What are circularly polarized light?

When the light vector has equal amplitudes of vibrations. It rotates along a circular path about the direction of propagation without any change in its magnitude. This light is called circularly polarized light.

24. What are elliptically polarized light?

When the light vectors has vibrations of unequal amplitudes it rotates in a elliptical path about the direction of propagation with change in its magnitude. Hence this light is called an elliptically polarized light.

25. Define Polariser and Analyser .

Polariser: A device which converts an unpolarised light into a plane polarized light is called a polarizer.

Analyser: A device which check whether the light is plane polarized or not is called as analyzer

Example: Nicol Prism.

26. Define optical activity.

The phenomenon or the property of rotating the plane of vibration by certain crystals or substances is known as optical activity.

27. Name some optically active substances.

Sugar crystals, turpentine, sodium chlorate , cinnabar etc.,

28. State the differences between ordinary ray and extra ordinary ray

Sl.No	Ordinary ray	Extra-ordinary ray
1	It obeys the laws of refraction	It does not obey the laws of refraction
2	It travels with a low velocity	It travels with a higher velocity than ordinary ray
3	Ordinary ray is stationary	Extra-ordinary ray revolves around the ordinary ray

29. What are dextro and laevo rotatory crystals?

The substances which rotate the plane of vibration to the right or clockwise direction are called dextro-rotatory or right handed rotation. The substances that rotate the plane of vibration to left or anticlockwise direction are called as laevo rotatory crystals.

30. Define a Wave plate

Wave plate is nothing a transparent plate made of uniaxial crystals like quartz or calcite cut parallel to the optic axis. When light falls on a wave plate at 90° to the optic axis it splits up into two rays

- a. Ordinary ray and
- b. Extra-ordinary ray

31. List out the uses of Quarter and Half wave plate**Quarter wave plate:**

1. It is used to convert a plane polarized light to circularly polarized light and vice versa
2. It is also used to convert a plane polarized light to elliptically polarized light and vice versa.

Half wave plate:

1. It is capable of rotating the plane polarized light through 90°

32. What is the use of half shade in Laurent's Half Shade Polarimeter.

The half shade serves the purpose of dividing the field of view in two halves. When the analyzing nicol is slightly rotated from the position of equal brightness a marked change in the intensity of two halves is observed.

PROBLEMS:

1. In the Michelson Interferometer arrangement, if one of the mirrors is moved by 0.04mm, 125 fringes cross the field of view. Calculate the wavelength of light used

Given data:

$$d = 0.04\text{mm} = 0.04 \times 10^{-3}\text{m}$$

$$n = 125$$

Solution:

Formula: Wavelength $\lambda = 2d/n$

$$= \frac{2 \times 0.04 \times 10^{-3}}{125}$$

$$= 6.4 \times 10^{-7}\text{m}$$

$$= 6400\text{\AA}$$

$$= 6400\text{\AA}$$

Wavelength of light used = 6400Å

2. Calculate the least thickness of (i) A Quarter wave plate and (ii) Half wave plate for a wavelength of 6000Å°. Given $\mu_e = 1.553$ and $\mu_o = 1.544$. (May 2017)

Solution: since $\mu_e > \mu_o$ the crystal is a positive uniaxial crystal

Formulae:

(i) Minimum thickness of quarter wave plate

$$t = \lambda / 4 (\mu_e - \mu_o)$$

$$t = 6000 \times 10^{-10} / 4(1.553 - 1.544) = 1.667 \times 10^{-5}\text{m}$$

(ii) Minimum thickness of Half wave plate

$$t = \lambda / 2 (\mu_e - \mu_o)$$

$$t = 6000 \times 10^{-10} / 2(1.553 - 1.544) = 3.33 \times 10^{-5}\text{m}$$

3. An airwedge experiment is used to find the thickness of a thinsheet of paper. A monochromatic light of wavelength 5893Å is falling normally on the air wedge of length 4.6X10⁻²m. If the thickness of 5 fringes is measured to be 4X10⁻³m. Calculate the thickness of the paper (Jan 2016)

Given data:

$$\lambda = 5893\text{\AA} = 5893 \times 10^{-10}\text{m}$$

$$d = 4.6 \times 10^{-2}\text{m}$$

$$5\text{fringes} = 4 \times 10^{-3}\text{m}$$

$$\beta = 0.8 \times 10^{-3}\text{m}$$

Solution:

Formula: $t = \lambda / 2 \beta$

$$= \frac{5893 \times 10^{-10} \times 4.6 \times 10^{-2}}{2 \times 0.8 \times 10^{-3}}$$

$$= 1.694 \times 10^{-5}\text{m}$$

$$= 1.694 \times 10^{-5}\text{m}$$

Pondicherry University PART B Questions

1. Describe the construction and working of Michelson's interferometer. How this can be used for measuring the wavelength of monochromatic light? (Jan 2010, Jan 2011, Jan 2013, Jan 2014, 2016, April/May 2011, May 2012, May 2017)

Answer: Refer Page No:7 to 11

2. Deduce an expression for air-wedge to find the thickness of a thin sheet of paper or wire. (Jan 2014)

Answer: Refer Page No:2 to 6

3. Explain briefly the resolving power of grating and prism (Jan 2014, April/May 2011)

Answer : Refer Page No:19 to22

4. (a) Explain Quarter and Half wave plates in detail. (Jan 2014)

Answer : Refer Page No:26 to 28

- (b) Write a short note on Dispersive power of grating.

Answer: Refer Page No:22

5. Briefly explain the following: (Jan 2013, 2016)

- (a) Antireflection coating (3)

Answer: Refer Page No:11to 13

- (b) Optical activity. (Answer: Refer Page No:28)

- (c) Explain double refraction using Huygen's theory (May 2017) (Answer: Refer Page No:25 to 26)

6. Deduce an expression for Air Wedge to find the thickness of a thin sheet of paper, and hence explain an experimental method to find the thickness of a thin sheet of a paper using Air Wedge Experiment? (April 2010, May 2011)

Answer: Refer Page No:2. to 6

7. Explain the term diffraction grating? (4) (May 2013)

Answer: Refer Page No:15 to 17

8. With neat diagram explain the construction and working of Laurent's half shade Polarimeter and explain the method to determine the specific rotation of sugar solution.(May 2017)

Answer: Refer Page No:29 to 32